Python Lectureflow

Module-1) SE - Overview of IT Industry	5

- Session 1 - Welcome to the IT World Explain careers in IT (developer, tester, designer, data analyst, AI engineer). Discuss product vs service companies; real-world examples (Google, Zoho, TCS). Overview of how teams collaborate using GitHub, Slack, Jira. Trainer Demo: Show an open-source GitHub project (e.g., “Awesome-Java”) and walk through contributors’ roles. AI Tool Tip: Ask ChatGPT to list top emerging tech jobs (to discuss in class).
- Session 2: How Software Works Define software application and its components. Explain software architecture layers: UI ? Business Logic ? Database. Compare desktop, web, and mobile apps. Explain client-server communication with example (browser request ? backend ? database ? response). Trainer Demo: Use Draw.io or Lucidchart AI to visually show data flow (e.g., for “Instagram feed refresh”).
- Session 3: Internet & Networking Basics Internet Fundamentals: How computers communicate. Protocols: Define HTTP, HTTPS, FTP, SMTP (overview only, 5 mins each). Domain & DNS: Explain how domain names map to IP addresses. Demonstrate using nslookup google.com. Explain DNS hierarchy: root, TLD, domain, subdomain. IP Addressing: What is an IP (IPv4 vs IPv6)? HTTP Lifecycle: Request ? Response ? Status Codes (focus on 200, 404, 500). Trainer Demo: Use browser DevTools ? Net
- Session 4: Software Development Lifecycle (SDLC) Explain SDLC phases: Requirement Design Development Testing Deployment Maintenance. Discuss Agile vs Waterfall, Scrum roles (PO, Scrum Master, Dev Team). Relate each phase to a real-world app (e.g., “Spotify new feature rollout”). Trainer Demo: Show a Jira board mock-up with stories in “To Do / In Progress / Done.”
- Session 5: Git & Version Control Why version control is needed. Git concepts: Repository, Commit, Branch, Merge, Push, Pull. GitHub overview: Forks, Pull Requests, Collaboration. Trainer Demo: Initialize a local Git repo (git init, git add ., git commit) Push to GitHub. AI Integration: Use GitHub Copilot Chat to explain git errors in real-time. Depth: Students should confidently push their first small project to GitHub.

Module-2) SE- Introduction to Web Technologies - HTML and CSS	12

- Session 1: Introduction to Web Design Coverage: How browsers render HTML + CSS. Difference between front-end & back-end. Developer tools overview (Chrome DevTools). Demo: Inspect elements on Instagram home page. Depth: Conceptual understanding of DOM structure.
- Session 2: HTML Page Structure Coverage: HTML boilerplate (<!DOCTYPE>, <html>, <head>, <body>). Tags: Headings, paragraphs, lists, anchors, images, tables. Demo: Create personal “About Me” page.
- Session 3: Forms & Inputs Coverage: Form elements (input, select, textarea), labels, validation attributes. Demo: Registration form for Social Bio Link App. Depth: Show POST request preview in DevTools Network tab.
- Session 4: Semantic HTML 5 Coverage: Semantic tags (header, nav, article, footer). Demo: Rebuild a YouTube-style page layout.

- Session 5: Introduction to CSS 3 Coverage: Inline, Internal, External styles; Selectors, Colors, Font families. Demo: Color-theme switch between light and dark mode.
- Session 6: Box Model & Positioning Coverage: Margin, Padding, Border, Display properties, Position (relative, absolute, fixed). Demo: Create a profile-card layout.
- Session 7: Flexbox & Grid Coverage: CSS Flexbox (container, justify, align) and Grid (template columns & rows). Demo: Responsive gallery of social icons.
- Session 8: Responsive Design Coverage: Media queries, Viewport, Breakpoints. Demo: Make the profile card mobile-friendly.
- Session 9: Bootstrap Basics Coverage: Bootstrap Grid System, Containers, Rows, Columns. Demo: 2-column layout with profile image and link list.
- Session 10: Bootstrap Components Coverage: Navbar, Buttons, Cards, Carousel, Forms. Demo: Navbar for Social Bio Link App.
- Session 11: Bootstrap Utilities & Customization Coverage: Spacing, Typography, Colors, Custom CSS overrides. Demo: Theme switcher using Bootstrap classes.
- Session 12: Integration Review Coverage: Integrate HTML + CSS + Bootstrap for a cohesive layout. Demo: Static mock-up of Bio Link landing page.
- Mini Project 6 – “Social Bio Link Page – UI Design” Objective: Build a personal bio link page (links to Instagram, LinkedIn, Spotify, etc.) with responsive design. Trainer Notes: Guide students to implement a form that accepts link inputs and shows preview cards.

Module 3) SE - Introduction to Programming	**15**

- Session 1: What is Programming? Topics: Define “program” and “programming language.” Explain compilers and interpreters. Discuss importance of algorithms and flowcharts. Trainer Activity: Draw a flowchart for “Ordering food online.”
- Session 2: Writing Your First C Program - Topics: Setting Up Environment -Installing a C Compiler (e.g., GCC), Choosing an IDE (DevC++. VS Code, Codeblocks, etc) Basic structure of C program (header, main, return). Compile and run using GCC or VS Code terminal. Example: “Hello, TOPS Technologies!”
- Session 3: Data Types and Variables Topics: Data types (int, float, char, double). Variable naming rules. Constants, literals. Example: Store and print student’s name, age, marks.
- Session 4: Operators and Expressions Topics: Arithmetic, Logical, Relational, Increment/Decrement. Operator precedence. Example: Simple calculator for addition, subtraction, etc.
- Session 5: Conditional Statements Topics: If, Else If Ladder, Nested If, Switch Case. Example: “Study Mood Bot” – takes user’s input and prints motivation message
- Session 6: Loops (For, While, Do-While) Topics: Difference between entry-controlled and exit-controlled loops. Use cases (e.g., menu-driven programs). Example: Countdown timer using loops.
- Session 7: Nested Loops & Pattern Printing Topics: Using nested loops for grids or triangle patterns. Example: Star pyramid pattern.
- Session 8: Functions Topics: Declaration, Definition, Calling. Pass-by-value vs pass-by-reference.
- Session 9: Arrays Topics: 1D & 2D Arrays, Array Traversal. Example: Record 7-day screen time & calculate average.
- Session 10: Strings Topics: Declaring and handling strings. Common functions: strlen(), strcpy(), strcmp(). Example: Create a username from name input.

- Session 11: Pointers Topics: Pointer declaration, dereferencing, pointer arithmetic. Example: Swap numbers using pointers. Trainer Demo: Use memory diagram to explain address referencing.
- Session 12: Structures Topics: Declaring, initializing structures. Nested structures. Example: Store student details (name, marks, grade).
- Session 13: File Handling Topics: Open, write, read, close files. Modes: r, w, a. Example: Write and read “daily goals.txt.”
- Session 14: Debugging & Best Practices Topics: Common syntax/runtime errors. Importance of commenting & indentation.
- Mini Project 1: “Student Productivity Tracker” Objective: Combine loops, arrays, and file handling to build a console-based app that logs daily study hours and generates a weekly progress report. Trainer Coverage: Guide students to design menu-driven interface. Discuss file I/O for persistence.

Module-4) Introduction to OOPS Programming	**8**

- Session 1: Procedural vs Object Thinking Topics: Limitations of C (global data, no encapsulation). Why OOP solves these. Example: Compare “Task List” written in C vs OOP style.
- Session 2: Classes and Objects Topics: Define class & object, syntax, member functions. Example: Class Note (title, date, isDone).
- Session 3: Constructors & Destructors Topics: Default, Parameterized, Copy constructors. Destructor lifecycle. Example: Auto-save user data upon exit.
- Session 4: Inheritance Topics: Single, Multilevel, Hierarchical inheritance. Example: ContentCreator ? YouTuber / Podcaster.
- Session 5: Polymorphism Topics: Compile-time (overloading) vs Runtime (overriding). Example: uploadContent() works differently for YouTube vs Instagram.
- Session 6: Encapsulation & Abstraction Topics: Access modifiers: public, private, protected. Abstract classes & pure virtual functions (conceptual). Example: Class Platform with abstract upload() method.
- Session 7: File Handling Topics: Streams (ifstream, ofstream), reading/writing files. Example: Save and retrieve creator analytics data.
- Mini Project 2: “Creator Dashboard Lite” Objective: Create a console app for digital creators to manage content ideas. Trainer Guidelines: Implement Content class (title, platform, views, status). Allow user to add, edit, delete content items. Store data using file handling.

Module-5) SE - Database Management – SQL & PL/SQL	**9**

- Session 1: Introduction to Databases Trainer Coverage: What is a database, DBMS, RDBMS (with examples). Difference between MySQL, PostgreSQL, Oracle, SQLite. Tables, rows, columns, and relationships. Demo: Show data stored in MySQL Workbench.
- Session 2: SQL Basics & Commands Trainer Coverage: SQL syntax rules. DDL commands: CREATE, ALTER, DROP. Creating basic users and expenses tables. Demo: Execute commands in MySQL CLI / Workbench.
- Session 3: DML Operations Trainer Coverage: Insert, Update, Delete records. WHERE and LIMIT usage. Example: Insert 5 sample users and edit their data.

- Session 4: Data Retrieval with SELECT Trainer Coverage: SELECT queries, filtering, sorting, aliases. Aggregate functions (SUM, AVG, COUNT). Demo: Generate “total expenses” by category.
- Session 5: Joins and Relationships Trainer Coverage: Foreign keys and primary keys. Inner Join, Left Join, Right Join. Demo: Link users with their transactions table.
- Session 6: Grouping and Subqueries Trainer Coverage: GROUP BY, HAVING. Subqueries and nested queries. Example: “Show users whose average daily spend > ?500.”
- Session 7: Constraints & Views Trainer Coverage: Constraints: NOT NULL, UNIQUE, PRIMARY KEY, FOREIGN KEY. Creating and querying VIEWS. Example: Create “ActiveUserView” showing users with >5 transactions.
- Session 8: Introduction to PL/SQL Trainer Coverage: PL/SQL block structure (DECLARE, BEGIN, EXCEPTION, END). Variables and cursors. Control structures (IF, LOOP). Demo: Simple stored procedure to calculate user’s total spend.
- Session 9: Transactions & Error Handling Trainer Coverage: COMMIT, ROLLBACK, SAVEPOINT. Triggers: use case “Auto update user’s balance after expense entry.” AI Integration: Use ChatGPT to simulate test triggers and rollback scenarios.
- Mini Project 3 – “Expense Tracker DB” Description: Design the backend for a Personal Expense Tracker App. Create users, transactions, and categories tables. Write SQL scripts for CRUD operations. Add a stored procedure for monthly summary generation. Trainer Deliverables: Students should submit .sql file with queries. Test queries via MySQL Workbench live.

Module 6) Python - Fundamentals of python language	**12**

- Session 1 - Introduction to Python What is Python? Why Python? Use cases Installing Python & VS Code Running first Python script Python code execution model
- Session 2 - Variables & Data Types Creating variables Naming rules Primitive & non-primitive types type(), id() Examples: ? Create variables for a student profile.
- Session 3 - Input, Output, Typecasting input() & print() int(), float(), str() conversions Escape characters Examples: Basic calculator.
- Session 4 - Conditional Statements if, else, elif Nested if Examples: ? Grade calculation ? Blood donation eligibility
- Session 5 - Loops (for) for loop basics Looping through strings, lists Examples: ? Print fruit names ? Count characters in string
- Session 6 - Loops (while) + Patterns while loop Infinite loop cases Star patterns
- Session 7 - Break, Continue, Pass break usage continue usage pass usage Examples: ? Skip “Banana” ? Break at “Banana”
- Session 8 - Strings Indexing Slicing String methods Examples: reverse string, substring extraction.
- Session 9 - Lists Creating lists Append, insert, remove, pop Sorting
- Session 10 - Tuples Tuple creation Accessing values immutability Examples: ? Tuple slicing ? Tuple concatenation
- Session 11 - Dictionaries Key/value pairs Adding/updating entries keys(), values(), items()
- Session 12 - Functions + Lambda Defining functions Arguments Return values Lambda functions + map, filter, reduce

Module 7) Python - Collections, functions and Modules in Python	12

- Session 1 - List Deep Dive Creating list with multi-data type Updating values Removing values
- Session 2 - List Iteration Patterns for loops on lists range with lists
- Session 3 - List Transformations round(), sorted(), sort() zip() usage
- Session 4 - Tuples Advanced Creating mixed tuples Slicing rules Tuple ? list conversion
- Session 5 - Dictionaries Deep Dive Updating values Nested dictionaries
- Session 6 - Converting Lists ? Dictionary Using zip() Using loops
- Session 7 - Counting Characters & Words freq dictionary word count program
- Session 8 - Functions Revisited Keyword & Positional arguments Default values
- Session 9 - Anonymous Functions Lambda with multiple expressions
- Session 10 - Modules (Custom) Creating student.py Importing module in main file
- Session 11 - math & random modules sqrt, ceil, floor randint, random
- Session 12 - Packages Creating package folder init.py usage
- session 13 - List comprehension - nested list
- session 14 - dynamic nested dictionary creation

Module 8) Python - Advance python programming	10

- Session 1 - File Handling Basics open(), read(), write() File modes: r, w, a
- Session 2 - File Pointer & Cursor Control tell(), seek() Reading line-by-line
- Session 3 - Exception Handling Basics try, except Multiple exceptions
- Session 4 - Custom Exceptions raise keyword Creating user-defined exception AI Tool: ChatGPT ? explain error tracebacks.
- Session 5 - Classes & Objects Class structure init method Creating objects
- Session 6 - Inheritance Single Multilevel Multiple Example: Family class inheritance.
- Session 7 - Polymorphism Method overloading (conceptual in Python) Method overriding
- Session 8 - Modules & Packages import module Creating own module math, random modules
- Session 9 - Regex (re module) match() search() findall()
- Session 10 - GUI Programming with Tkinter Window Labels Buttons Layout
- session11 : Database connecting with python - pymysql or mysql connectivity

Module 9) Python - DB and Python Framework	20

- SESSION 1 - Introduction to Django & Setup Topics What is Django? Why Django vs Flask Django features MVT Architecture Installing Django Creating first Django project Running Django server Trainer Activities Create project: doctorfinder_project Explore project structure
- SESSION 2 - Apps, Settings & URL Routing Topics Creating Django app Project vs App settings.py, urls.py, wsgi.py Mapping URL ? View HttpResponse basics Activities Create app: accounts Add it to INSTALLED_APPS Create first view and URL
- SESSION 3 - Templates & Static Files Topics Templates folder structure Template inheritance (base.html) Loading static files (CSS, JS) Using template tags Activities Build base.html layout Add Bootstrap via CDN

- SESSION 4 - Django Models & Database Connectivity Topics Creating models Model fields SQLite ? MySQL migration Migrations: makemigrations, migrate ORM basics Activities Create model: Doctor, Specialization Connect Django to MySQL using pymysql
- SESSION 5 - Django Admin Customization Topics Enabling admin Adding models to admin Custom list display Search fields, filters Activities Customize admin for Doctor model Include filters for specialization
- SESSION 6 - Django Forms Basics Topics Django Form class GET vs POST CSRF token Handling form submission Activities Build “Add Doctor” form Display validation errors
- SESSION 7 - Django Model Forms Topics ModelForm benefits Automatic form generation Saving form data to model Activities Create DoctorForm Save doctor details from form
- SESSION 8 - User Authentication System Topics User model Signup Login Logout Password hashing Authentication backend Activities Build Register ? Login ? Logout Display logged-in username in navbar
- SESSION 9 - Email Sending in Django Topics SMTP setup Django email functions HTML email templates Activities Send welcome email on signup
- SESSION 10 - Forgot Password + Session Management Topics Django sessions Reset password with OTP link Session expiry Middleware basics Activities Implement forgot password Store OTP in session
- SESSION 11 - Working With Media Files & User Profile Topics MEDIA_ROOT and MEDIA_URL Upload profile picture Serving uploaded files Activities Profile edit page (update photo, name, phone)
- SESSION 12 - CRUD Operations Topics Create, Read, Update, Delete Editing using ModelForm Delete confirmation page Activities CRUD for Doctor: Add ? View ? Edit ? Delete
- SESSION 13 - CRUD Using AJAX Topics AJAX introduction Fetch API Sending JSON request Handling JSON responses Activities Implement delete doctor using AJAX Show success message dynamically
- SESSION 14 - Django ORM: QuerySet Deep Dive Topics filter(), exclude(), annotate() select_related(), prefetch_related() Aggregations Q() objects Activities Search doctors by multiple filters Paginate doctor listing
- SESSION 15 - Django Authentication (Advanced) Topics Login required decorator Role-based access (doctor vs patient) Permissions Groups Activities Restrict admin pages Create role-based dashboards
- SESSION 16 - Django Integrations Topics Social Login (Google) Email APIs (Mailchimp / Mailgun) SMS APIs (Twilio) Activities Integrate Google Login for patient Send OTP via SMS
- SESSION 17 - Payment Gateway Integration Topics Paytm/Stripe/PayPal integration Payment initiation Transaction callback Activities Create “Book Appointment” ? “Pay Now” flow
- SESSION 18 - Maps & Geolocation API Topics Google Maps API Geocoding (address ? latitude, longitude) Embedding map in Django Activities Show doctor clinic location on map Search doctors by distance
- SESSION 19 - GitHub Deployment & PythonAnywhere Hosting Topics Push Django project to GitHub PythonAnywhere deployment Static & media files config Database migration on server Activities Fully deploy Doctor Finder project Test live routes
- SESSION 20 - Capstone: Doctor Finder Project (Integration & Cleanup) Includes Registration, Login, OTP, Forgot Password Profile Management Search Doctors Filters (specialization, location)

Appointment booking Google Maps integration Payment gateway Admin management Live deployment Deliverables Final code on GitHub Final deployment URL Project documentation

Module 10)Django REST Framework (REST API Development)	**6**

- SESSION 1 - Introduction to REST & DRF Setup Topics Covered What is an API? What is a REST API? (Principles: stateless, resource-based, CRUD) JSON vs XML Difference between Django & Django REST Framework Installing DRF Adding rest_framework to INSTALLED_APPS Creating a new Django app: api Serializer vs ModelSerializer overview First test API endpoint Trainer Demonstration Create project folder /api Create simple “Hello API” endpoint using APIView or function-base
- SESSION 2 - Serializers + CRUD API Development Topics Covered Creating models for API Writing serializers (Serializer + ModelSerializer) Creating CRUD endpoints: Create (POST) Read (GET) Update (PUT/PATCH) Delete (DELETE) Using APIView / GenericAPIView / Mixins JSON responses Status codes (HTTP_200, 201, 404, 400) Trainer Demonstration Build CRUD API for Doctor model: /api/doctors/ /api/doctors/<id>/
- SESSION 3 - ViewSets, Routers & Pagination Topics Covered ViewSets (ModelViewSet) Routers & automatically generated URLs Default Router vs SimpleRouter Pagination types: PageNumberPagination LimitOffsetPagination CursorPagination Ordering and filtering basics Trainer Demonstration Convert CRUD API to ModelViewSet Implement pagination /api/doctors/?page=2 Practice Add ordering by name or specialization.
- SESSION 4 - Authentication & Permissions Topics Covered BasicAuthentication TokenAuthentication SessionAuthentication Permissions: AllowAny IsAuthenticated IsAdminUser Custom Permissions JWT Overview (Optional light introduction) Trainer Demonstration Protect /api/doctors/ using authentication Generate auth token for user (using rest_framework.authtoken) AI Integration ChatGPT ? Explain “Token vs Session Auth” for interview prep.
- SESSION 5 - Third-Party API Integrations Topics Covered Integrating Popular External APIs: OpenWeatherMap API Google Maps Geocoding API REST Countries API GitHub API Twitter API (theoretical due to OAuth limits) Trainer Demonstration Create endpoint /api/weather/<city>/ DRF view calls external weather API Show JSON response to frontend AI Tool Integration Postman AI ? test API endpoints ChatGPT ? generate Python requests code snippet.
- SESSION 6 - Advanced Features + Deployment Topics Covered Email Sending API (MailChimp, Mailgun) SMS API (Twilio) Payment APIs (Stripe/PayPal) Social Authentication (Google Login) Custom Response Formatting Exception Handling in DRF Deploying REST API on PythonAnywhere Versioning your API (v1, v2) Trainer Demonstration Integrate sample Twilio SMS API (demo level) Deploy REST API live Test endpoints using Postman

Module 11) JavaScript Essentials And Advanced	**20**

- Session 1: Introduction to JavaScript & Environment Setup Coverage Depth: What JavaScript is (client-side vs server-side). How browsers interpret JS and connect it with HTML via <script> tag. Running JS in console (browser + VS Code Live Server). Demo: Add alert("Welcome to your Bio-Link App!") inside an HTML file. Learning Goal: Students understand how JS executes and connects to DOM.

- Session 2: Variables & Data Types Coverage Depth: var, let, const-scope & reassignment. Primitive vs Reference types. Dynamic typing in JS. Demo: Create variables: let userName="Aarya", let followers=1500. Display in console. Link to App: Represent profile details in JS variables.
- Session 3: Operators & Conditions Coverage Depth: Arithmetic, Logical, Comparison, and Ternary operators. Truthy/Falsy values concept. Demo: If followers > 1000 ? show “Verified Creator ??”. Trainer Tip: Relate conditional checks to real user-verification logic (like Instagram).
- Session 4: Loops & Iteration Coverage Depth: For, While, Do-While, For-of, ForEach. Difference between indexed and iterable loops. Demo: Loop through an array of social links and print each in console. Learning Goal: Understand repetition and dynamic rendering concept.
- Session 5: Functions & Reusability Coverage Depth: Function declaration vs expression vs arrow function. Parameters and return values. Demo: Function createButton(label,color) that returns a styled HTML button.
- Session 6: DOM Manipulation – Selecting Elements Coverage Depth: getElementById, querySelector, innerHTML, style. Reading & updating content dynamically. Demo: Add new link element dynamically from input form. Trainer Note: Show real-time DOM change in browser.
- Session 7: DOM Events & Interactivity Coverage Depth: Event types: onclick, onchange, addEventListener. Event object basics. Demo: “Add Link” button creates new card instantly. Depth: Explain event flow concept (intro only).
- Session 8: Arrays & Objects – Data Structures Coverage Depth: Array creation, methods (push, pop, splice). Object literals and nested objects. Demo: let links=[{title:"Instagram",url:"..."},...] Loop to display all cards. Learning Goal: Represent app data as structured collections.
- Session 9: JSON & LocalStorage (Persistence Basics) Coverage Depth: Converting data: JSON.stringify() ? JSON.parse(). Using localStorage.setItem/getItem/removeItem. Demo: Save links to localStorage; reload page = data persists. Trainer Note: Relate this to database persistence later (JDBC concept preview).
- Session 10: ES6 Syntax & Modern Features Coverage Depth: Template literals, default params, destructuring, spread/rest. Arrow functions recap. Demo: console.log(\Hello ${userName}, you have ${followers} followers!`)`. Depth: Use modern syntax in all subsequent examples.
- Session 11: Asynchronous JS – Callbacks, Promises, Async/Await (Part 1) Coverage Depth: Blocking vs non-blocking code. What are callbacks? Creating and resolving Promises. Demo: Simulate loading message using setTimeout.
- Session 14: Modular JavaScript (Imports & Exports) Coverage Depth: File modularization with export & import. Default vs named exports. Demo: Move link-creation functions to utils.js and import in main file. Depth: Explain benefit for scalability (pre-React architecture).
- Session 12: Asynchronous JS (Part 2) – Fetch API Coverage Depth: fetch() GET & POST. Chaining Promises & handling responses. Parsing JSON responses. Demo: Fetch mock user data from https://jsonplaceholder.typicode.com/users. Depth: Show network tab request ? response cycle.
- Session 15: Classes & OOP in JS Coverage Depth: Declaring classes, constructors, methods. Creating instances. Demo: class UserProfile { constructor(name, followers){...} } Add method displayProfile(). Trainer Note: Show parallel with Java classes ? OOP bridge for Phase 4.

- Session 13: Error Handling in JS Coverage Depth: try…catch…finally, custom errors, throw. Importance of defensive coding. Demo: If URL input invalid ? throw custom Error (“Invalid URL”). AI Integration: ChatGPT explains error stack trace for student practice.
- Session 16: Advanced Array Methods Coverage Depth: map(), filter(), reduce(), find(), some(), every(). Real use cases for transformation. Demo: Filter active links (isActive:true), calculate total clicks. AI Tool: Ask ChatGPT to suggest one-liner map-reduce alternatives.
- Session 17: Event Delegation & Dynamic UI Updates Coverage Depth: Event bubbling vs capturing. Delegating events for dynamic elements. Demo: Click “X” icon ? remove card without re-binding listeners. Depth: Students must grasp efficiency benefit.
- Session 18: Debugging & Developer Tools Coverage Depth: Console methods (log, warn, error, table). Setting breakpoints in Sources tab. Understanding stack traces. Demo: Debug addLink() and inspect variables. AI Tip: Ask ChatGPT to explain error messages from console.
- Session 19: Mini-Project Preparation & Integration Coverage Depth: Combine HTML, CSS, and JS into one interactive application. Plan modules: Form ? Data Storage ? Display ? Edit/Delete. Activity: Sketch architecture on whiteboard (show how data flows between functions). Trainer Goal: Help students map concepts to app features clearly.
- Session 20: Integration Testing & Polish Coverage Depth: Manual QA: test each button and form field. Common bugs & fixes (null values, double adds). Basic UI validation patterns. Demo: Final run-through of Social Bio Link App.
- Mini Project 7 – Interactive Social Bio Link Builder Duration: 2 – 3 hours (integrated across Sessions 19–20) Description: Create an interactive web app that lets users: Add, edit, and delete social links. Persist links using LocalStorage. Apply light/dark mode toggle. Validate URLs and display error messages. Trainer Focus: Emphasize modular coding and event-driven logic. Reinforce reusability (functions ? classes). Highlight how data persistence today (LocalStorage) will evolve to real database

Module-12) React - Fundamentals	**14**

- Session 1 – Introduction to React Topics to Cover: What is React? Why React (Virtual DOM, Components). create-react-app setup. Demo: Run first React app and modify default component.
- Session 2 – JSX & Components Topics to Cover: JSX syntax and rules. Function vs Class components. Demo: Render “Hello {username}” component.
- Session 3 – Props Topics to Cover: Passing data between components. Default props and prop types. Demo: Reusable “Card” component with props.
- Session 4 – State & useState Hook Topics to Cover: State management basics. Updating state. Demo: Counter increment/decrement app.
- Session 5 – Event Handling Topics to Cover: Handling click, input, form events. Demo: Interactive form capturing user data.
- Session 6 – Lists & Conditional Rendering Topics to Cover: Rendering arrays with map(). Conditional rendering (if, ternary). Demo: Render dynamic product list.
- Session 7 – useEffect Hook Topics to Cover: Component lifecycle and side effects. Demo: Simulate API fetch on mount.
- Session 8 – useRef & Controlled Forms Topics to Cover: Managing DOM elements using refs. Demo: Focus input field after submit.

- Session 9 – Routing Topics to Cover: React Router setup. Routes, NavLink, BrowserRouter. Demo: Multi-page navigation (Home, About, Contact).
- Session 10 – Context API Topics to Cover: Global state management. useContext hook. Demo: Theme toggle (dark/light mode).
- Session 11 – Axios & API Integration Topics to Cover: GET and POST using Axios. Demo: Fetch product list from mock API.
- Session 12 – Error Handling Topics to Cover: Try/catch in API calls. Demo: Display “Error loading data” message.
- Session 13 – Deployment Topics to Cover: npm run build, Netlify deployment. Demo: Publish React SPA online.
- Session 14 – Mini Project: React Portfolio Builder Goal: Build portfolio website with reusable components. AI Add-on: Use Copilot to suggest component structure.

Module 13) React Advanced (Hooks, Firebase, Redux)	20

- Session 1 – Introduction to Advanced React Concepts Topics to Cover: Review: props, state, useEffect, component hierarchy. Why scalability matters (component reusability, maintainability). Introduction to advanced hooks (custom hooks, reducer hooks). Depth: Explain problems with prop drilling and redundant code in large apps. Practical Demo: Show an example of repetitive API call logic to motivate custom hooks.
- Session 2 – Custom Hooks (Part 1) Topics to Cover: What is a custom hook and naming conventions (useSomething). Converting logic from components into hooks. Rules of hooks (only call at top level). Demo: Build useFetchData() hook for fetching API data. Depth: Explain how custom hooks promote reusability across components.
- Session 3 – Custom Hooks (Part 2) Topics to Cover: Passing parameters to custom hooks. Managing multiple useEffects inside a custom hook. Returning data and loading/error states. Demo: Enhance useFetchData() to include loading spinner and error handling.
- Session 4 – useReducer Hook (Part 1) Topics to Cover: Concept of reducers (state + action ? new state). useReducer() syntax and initialization. Demo: Build counter example with increment/decrement/reset using useReducer. Depth: Explain the difference between useState vs useReducer for complex state logic.
- Session 5 – useReducer Hook (Part 2) Topics to Cover: Multiple action handling (switch case). Dispatching from child components. Demo: To-Do app using reducer for add, edit, delete actions.
- Session 6 – useMemo & useCallback Hooks Topics to Cover: React re-rendering and performance bottlenecks. Memoization with useMemo(). Function memoization with useCallback(). Demo: Create large list rendering with and without memoization; compare performance.
- Session 7 – Context API Deep Dive (Part 1) Topics to Cover: What is global state and why needed. Creating Context and Provider. Passing state to deeply nested components. Demo: Create ThemeContext for global dark/light mode. Depth: Explain prop drilling problem and how Context solves it.
- Session 8 – Context API Deep Dive (Part 2) Topics to Cover: Combining Context with Reducer. Nested context providers. Avoiding unnecessary re-renders using memoization. Demo: User authentication context using reducer + context.

- Session 9 – Firebase Introduction Topics to Cover: What is Firebase and its role (backend as a service). Firebase setup: project creation, SDK integration. Environment variable management for Firebase keys. Demo: Initialize Firebase project and connect to React app. Depth: Explain benefits of serverless integration for front-end developers.
- Session 10 – Firebase Authentication (Part 1) Topics to Cover: Sign-up, login with email/password. Firebase Auth API overview. Demo: Build SignUp and Login components using Firebase methods.
- Session 11 – Firebase Authentication (Part 2) Topics to Cover: Persistent authentication with onAuthStateChanged(). Logout and user session handling. Demo: Implement logout functionality with user greeting in navbar. Depth: Emphasize secure handling of user session states.
- Session 12 – Firestore CRUD Operations (Part 1) Topics to Cover: Firestore introduction (collections, documents). Add and read data. Demo: Add and fetch tasks in a "Task Manager" app.
- Session 13 – Firestore CRUD Operations (Part 2) Topics to Cover: Update and delete documents. Real-time updates with onSnapshot(). Demo: Real-time task updates without refreshing. Depth: Explain how Firebase syncs UI state automatically.
- Session 14 – Redux Fundamentals Topics to Cover: Redux architecture: store, action, reducer, dispatch. Setting up Redux in a React project. Demo: Create simple counter using Redux store.
- Session 15 – Redux State Flow Topics to Cover: Connecting components with useSelector and useDispatch. Redux DevTools overview. Demo: Display and update tasks using Redux actions. Depth: Show difference between Context and Redux for large-scale apps.
- Session 16 – Async Operations in Redux (Thunk) Topics to Cover: Middleware concept. Async API calls using Redux Thunk. Demo: Fetch task list from Firebase using thunk action.
- Session 17 – Form Handling with Formik Topics to Cover: Controlled vs uncontrolled components. Installing and using Formik. Validation with Yup schema. Demo: Task creation form with validation. Depth: Highlight cleaner syntax and validation vs manual handling.
- Session 18 – Environment Variables & Configuration Topics to Cover: Environment management (.env.local). API key security. Conditional environments (dev vs prod). Demo: Secure Firebase credentials in environment files.
- Session 19 – Mini Project Build: "Task Manager App" (Part 1) Topics to Cover: Create base structure: authentication + CRUD logic. Redux integration for task management. Demo: Build and test authentication + task list UI. AI Tip: Use Copilot to assist in creating CRUD reducers and components.
- Session 20 – Mini Project Deployment & Review Topics to Cover: Firebase Hosting or Netlify deployment. Testing, debugging, performance optimization. Final code walkthrough. Demo: Deploy the Task Manager app live and share link. AI Integration: Ask ChatGPT: "Generate README.md for this React + Firebase project."

Module-14) GraphQL & Next.js	**10**

- Session 1 – Introduction to GraphQL Topics to Cover: What is GraphQL and how it differs from REST APIs. Key concepts: Schema, Query, Mutation, Resolver. Benefits: Fewer API calls, strongly typed queries. Depth: Students must understand how GraphQL allows frontends to request only the data they need. Practical Demo: Use the public GraphQL Pokémon API in GraphiQL Playground - run a simple query to fetch names and images.

- Session 2 – GraphQL Setup in React Topics to Cover: Setting up Apollo Client. Installing dependencies: @apollo/client and graphql. Connecting to GraphQL endpoint. Demo: Configure ApolloProvider and test a simple query to display user names. Depth: Explain the Apollo Provider's role as React Context for GraphQL data.
- Session 3 – Writing Queries and Displaying Data Topics to Cover: Basic useQuery hook usage. Fetching and rendering data dynamically. Handling loading and error states. Demo: Build "User Directory" fetching list of users with names and emails. Depth: Explain destructuring of { loading, error, data } and conditional rendering.
- Session 4 – Mutations (Add, Update, Delete) Topics to Cover: Writing GraphQL mutations. Using the useMutation hook in Apollo. Optimistic UI updates. Demo: Add a new user entry using mutation and update UI instantly. Depth: Show how mutation modifies backend and automatically re-renders the component.
- Session 5 – GraphQL Schema & Playground Practice Topics to Cover: Understanding schema structure and types. Custom queries and relationships. Using Apollo Sandbox/GraphiQL for testing. Demo: Test nested queries and learn introspection. Depth: Students must understand what type definitions mean (type User { id: ID! name: String! }).
- Session 6 – Introduction to Next.js Topics to Cover: What is Next.js, and why use it over Create React App. SSR (Server-Side Rendering) vs CSR (Client-Side Rendering). Next.js folder structure overview. Demo: Create first Next.js app using npx create-next-app. Depth: Explain how SSR improves SEO and load performance.
- Session 7 – Next.js Pages and Routing Topics to Cover: File-based routing: /pages directory. Dynamic routes [id].js. Navigation with Link component. Demo: Build 3-page blog app (Home, About, Post Detail). Depth: Clarify automatic routing and static path generation. AI Add-on:
- Session 9 – Authentication and Environment Variables Topics to Cover: Adding Firebase or NextAuth.js for authentication. Environment files in Next.js (.env.local). Protecting API routes. Demo: Add Google login to Next.js app. Depth: Teach handling protected routes with session validation.
- Session 8 – Data Fetching in Next.js Topics to Cover: getStaticProps, getServerSideProps, getStaticPaths. When to use each method. Combining with GraphQL queries. Demo: Fetch and render blog posts at build time using getStaticProps. Depth: Explain difference between SSR and SSG with diagrams.
- Session 10 – Mini Project: "AI-Powered Blog Platform" Goal: Build a fully functional blog platform integrating: Next.js for SSR + Routing GraphQL API for content Authentication (Firebase/NextAuth) OpenAI API for AI-generated article summaries

Module 15) AI Tools for Front-End Developers	**6**

- Session 1 – The Role of AI in Modern Web Design Topics to Cover: What is generative AI and its place in web development. Overview of AI tools: ChatGPT, Copilot, Firefly, Figma AI, Uizard. Ethical AI usage and prompt engineering basics. Depth: Students should understand how AI assists developers (not replaces them). Demo: Use ChatGPT to generate a sample landing page structure (HTML outline).
- Session 2 – Using ChatGPT & GitHub Copilot for Coding Topics to Cover: How to use ChatGPT for debugging, explaining errors, and refactoring. Installing and using Copilot inside VS

Code. Writing effective prompts for code generation. Depth: Emphasize that AI helps accelerate, not automate without review. Demo: Refactor a React component using Copilot auto-suggestions. AI Exercise: Ask ChatGPT: “Optimize this React component for cleaner code and reusable hooks.”
- Session 3 – Figma AI and Uizard for Design Automation Topics to Cover: Using Figma AI to generate wireframes. Uizard overview – text-to-design prototypes. Converting Figma designs to HTML/CSS. Demo: Type a text prompt in Uizard (“Create a fitness app homepage”) ? export design preview. Depth: Discuss design-to-code handoff workflows. AI Tip: Show Figma plugin “Figma to Code” in action for live export.
- Session 4 – AI Image & Asset Generation Topics to Cover: Using Adobe Firefly / Leonardo.ai for image creation. Generating icons, backgrounds, and hero images using prompts. Image optimization (size and format). Demo: Generate a “modern tech startup hero image” using Firefly ? import to project. Depth: Discuss prompt tone (lighting, color, realism). AI Integration: Show differences between Midjourney and Firefly results.
- Session 5 – SEO & Content with AI Topics to Cover: Creating meta titles, keywords, and alt text using ChatGPT. Generating blog posts and product descriptions. Avoiding plagiarism and bias. Demo: Generate SEO tags for a blog post in ChatGPT ? integrate into HTML <head>. Depth: Discuss importance of keyword clustering for organic reach. AI Tip: Ask: “Generate SEO title, description, and alt text for an e-commerce landing page.”
- Session 6 – Mini Project: “AI-Generated Landing Page” Goal: Build a fully AI-assisted website prototype. Steps: Use Uizard or Figma AI for layout. Generate hero images via Firefly. Write content (headlines, text) using ChatGPT. Code the layout in HTML/CSS/React. AI Integration: Copilot for code generation, ChatGPT for copywriting and bug explanation.

Module-16) Capstone Project	**10**

- Session 1 – Project Orientation & Idea Finalization Topics to Cover: Explain project scope and goals. Brainstorm ideas and finalize concept. Example Project Themes: AI Resume Builder (React + Firebase + OpenAI API) Smart E-Commerce Dashboard (Next.js + GraphQL) AI Blog Creator (React + Firebase + ChatGPT content) AI Integration: Use ChatGPT to brainstorm project features and flow diagrams.
- Session 2 – Wireframing & UI Design Topics to Cover: Low-fidelity vs high-fidelity wireframes. Figma AI / Uizard prototyping. Demo: Create a wireframe for homepage and dashboard UI. Depth: Discuss user journey mapping and layout hierarchy. AI Tip: Ask Figma AI: “Generate layout for job profile dashboard.”
- Session 3 – Environment Setup & Boilerplate Creation Topics to Cover: Next.js / React app setup. Folder structure for scalability. Environment variables for APIs. Demo: Create .env.local for OpenAI or Firebase keys. AI Integration: Ask ChatGPT: “Generate secure .env configuration for my project.”
- Session 4 – Component Development (Frontend) Topics to Cover: Navbar, Footer, Cards, Forms, Modals. Props and State flow. Demo: Build dynamic navigation + form component. Depth: Focus on reusable, modular component design. AI Tip: Use Copilot for scaffolding repetitive UI components.
- Session 5 – Backend Integration (Firebase or GraphQL) Topics to Cover: Setting up Firestore or GraphQL API. CRUD operations integration. Demo: Connect form to Firestore to store and fetch data. AI Integration: Ask ChatGPT: “Refactor this API call with async/await and better error

handling.”

- Session 6 – Authentication & State Management Topics to Cover: Firebase Auth / NextAuth.js. Redux or Context API for session management. Demo: Implement login + user dashboard with persisted session. Depth: Emphasize UI reactivity to login state.
- Session 7 – AI API Integration Topics to Cover: Using OpenAI API or Hugging Face API. Fetching AI-generated content dynamically. Demo: Generate blog summary or resume content from user input using OpenAI API. AI Add-on: Show how prompt tuning changes output quality.
- Session 8 – Responsive Design & Testing Topics to Cover: Testing on mobile, tablet, desktop using DevTools. Browser compatibility checks. Lighthouse performance optimization. Demo: Run Google Lighthouse and fix issues. AI Integration: Ask ChatGPT: “Suggest improvements for Lighthouse performance report.”
- Session 9 – Final Deployment Topics to Cover: Hosting options: Firebase Hosting, Netlify, or Vercel. Connecting custom domain. Demo: Deploy project and verify live link. Depth: Discuss CI/CD basics briefly for future learning.